

Vol 11 Chapter 1 — Bounded Hermitian Symmetric Domains

Keeper (author pass — deep math/physics revision)

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Chapter 1 — Bounded Hermitian Symmetric Domains

Legacy chapter (May 2026). Last accuracy-checked 2026-09-11 (Keeper, K1892): the presentation scan found no retired reading here, and the tiers in the front matter are the May tiers, not re-verified line by line. Where this chapter and the Spine (Curriculum/Spine_DIV5_QM_GR_SM/) disagree, the Spine is current; where the Spine and the theorem registry disagree, the registry wins.

Level 1 — one sentence

Bounded Hermitian symmetric domains — bounded open subsets of $\mathbb{C}^n$ realizing Hermitian symmetric spaces of non-compact type — are classified by Cartan (1926) into four classical series (Types I-IV) and two exceptional cases, with the BST substrate $D_{IV}^5 = SO_0(5, 2)/[SO(5) \times SO(2)]$ being the specific Type IV bounded domain of dimension 5.

Level 2 — graduate-physicist precision

1.1 Hermitian symmetric spaces

A Hermitian symmetric space is a Riemannian manifold M with: - An almost complex structure J (gives complex-manifold structure) - A Hermitian metric compatible with J - An involutive isometry at each point (symmetry)

Cartan classification (1926): Hermitian symmetric spaces of non-compact type split into: - Type I: $D_{p,q}^I = SU(p, q)/[S(U(p) \times U(q))]$ — generalized unit ball - Type II: $D_n^{II} = SO^*(2n)/U(n)$ — antisymmetric matrices - Type III: $D_n^{III} = Sp(n, \mathbb{R})/U(n)$ — Siegel upper half-space - Type IV: $D_n^{IV} = SO_0(n, 2)/[SO(n) \times SO(2)]$ — Lie ball / Cayley domain - Two exceptional: $E_{6,(-14)}$ and $E_{7,(-25)}$

1.2 Bounded realizations

Each non-compact Hermitian symmetric space admits a bounded realization as an open subset of $\mathbb{C}^d$ (where d is the complex dimension). The boundary is non-trivial (has interior structure — Shilov boundary, characteristic subboundary, etc.).

For Type IV D_n^{IV} : realized as

$$D_n^{IV} = \{z \in \mathbb{C}^n : |z^T z|^2 - 2z^* z + 1 > 0, |z^T z| < 1\}$$

(Lie ball realization). Complex dimension n .

1.3 D_{IV}^5 specifically

For BST: $n = 5$.

$D_{IV}^5 = SO_0(5, 2)/[SO(5) \times SO(2)]$, complex dim 5 = real dim 10.

Maximal compact subgroup $K = SO(5) \times SO(2)$, dim 10 + 1 = 11.

Group dim: $\dim SO_0(5, 2) = 21$.

Rank 2 (Cartan), with positive roots whose half-sum $\rho = (5/2, 3/2)$.

1.4 Shilov boundary

The Shilov boundary S of a bounded domain is the smallest closed subset on which every holomorphic function attains its maximum modulus.

For D_{IV}^5 : Shilov boundary is a specific 5-real-dimensional submanifold of the topological boundary. Holomorphic functions on the closure of D_{IV}^5 satisfy a maximum principle on S — boundary values determine the function.

Substrate reading (Casey's Saturday work): electron = "Shilov-boundary primitive cycle." The substrate's Shilov boundary structure organizes specific BST particle K-types.

1.5 Cartan parameters

The rank-2 Cartan subalgebra of $\mathfrak{so}(5, 2)$ gives the substrate's natural parameters. Half-sum of positive roots: $\rho = (5/2, 3/2)$.

These appear throughout BST mass formulas, Bergman normalization, and operator zoo Casimir eigenvalues.

1.6 Why D_{IV}^5 specifically (BST uniqueness)

BST identifies D_{IV}^5 as the unique substrate via convergent constraints (Strong-Uniqueness Theorem progress, Lyra Task #206): - Type IV is the only rank-2 series with no spinor constraint - $n = 5$ is the unique dimension where $N_c \rightarrow$

$C_2 = 6 =$ quadratic Casimir of (1,1) - $N_{\max} = 27 \cdot 5 + 2 = 137$ — substrate's natural pre- α value - ... 8 more substrate-uniqueness criteria documented in BST Strong-Uniqueness Theorem

1.7 K-audit anchors

- Wallach 1976 (L1 ESTABLISHED): unitary reps on bounded symmetric domains
- Vol 5 Ch 1: Bergman $H^2(D_{IV}^5)$
- Strong-Uniqueness Theorem (Lyra Task #206): D_{IV}^5 uniqueness criteria

Level 3 — 5th-grader accessibility

Hermitian symmetric spaces = nice complex manifolds with extra symmetry. Cartan (1926) classified all of them. Four infinite families (Types I-IV) + 2 exceptional. $D_{IV}^5 =$ Type IV, dimension 5 = $SO_0(5, 2)/[SO(5) \times SO(2)]$ — the BST substrate. Complex dim 5, real dim 10. Shilov boundary: special boundary where holomorphic functions take their max — substrate of electrons in BST. Why D_{IV}^5 specifically: 11+ uniqueness criteria converge on this single geometry (Strong-Uniqueness Theorem).

What comes next

Chapter 2 develops Bergman reproducing kernels.

Where to look this up

- Helgason, *Differential Geometry, Lie Groups, and Symmetric Spaces*
- Faraut and Koranyi 1990
- BST: Strong-Uniqueness Theorem; Vol 5 Ch 1